HARSHALI SURESH BADKI

Lohegaon, Pune | 8999487012 | harshalibadki875@gmail.com
https://github.com/harshalibadki875 | https://www.linkedin.com/in/harshali-badki-a89780286/

Motivated and dedicated Aeronautical Engineering student with hands-on experience in Solidworks and CATIA V5 software for 3D modeling and assembly. Interested in aircraft structures, UAV design, and structural analysis. Enthusiastic to utilize technical skills in actual aerospace applications and gain industry experience.

INTERNSHIP

IIT Kanpur
Intern – Flight Laboratory **Jul 2025 – Aug 2025**

- Gained exposure to introduction of flight experiments and aircraft performance concepts.
- Understood basic flight testing procedures and experimental methodologies.
- Learned practical aspects of aerodynamics and aircraft behavior during flight.
- Observed and analyzed experimental data related to flight performance.

INNOVATIVE UAV & CAD PROJECTS

Tilt rotor RC Plane — Designed and successfully flight-tested a tilt rotor UAV with smooth VTOL transition and stable aerodynamic performance.

Drone Design — CATIA V5 — Created a lightweight UAV model in CATIA V5. Collaborated with faculty to verify design feasibility and manufacturability.

Landing Gear Design — Solidworks — Engineered a UAV landing gear system in SolidWorks, validating load distribution. Optimized materials for strength and weight efficiency.

Turbo Fan Engine Modeling — Solidworks — Developed a detailed CAD model of a turbofan engine advised by the GE CF6 using SolidWorks. Focused on accurate component integration, aerodynamic flow modeling, and mechanical detailing to replicate real-world engine functionality.

Cessna 172 Aircraft Design — Solidworks — Modeled and assembled a Cessna 172-inspired aircraft referencing IIT Kanpur's design framework. Experienced the aircraft's initial flight demonstration, gaining insights into aerodynamic stability, control behavior, and real-world flight characteristics.

CAD & Assembly Design Projects — Produced high-precision 3D models, assemblies for aerospace and mechanical components using CATIA and SolidWorks. Ensured design accuracy, manufacturability, and adherence to engineering standards.

EDUCATION

Ajeenkya DY Patil University, Pune | 2023-2027
B.Tech in Aeronautical Engineering – 9.2 cgpa

Lokmanya Tilak Vidyalaya | 2021–2023
Higher Secondary (12th) – 72%

Learners' International School | 2019–2021
Secondary (10th) – 93%

SKILLS

- 3D CAD & Modeling: CATIA V5, Solidworks, AutoCAD
- Engineering Documentation: 2D drawings, technical assemblies
- MS Office: Word, Excel, PowerPoint
- Python (Basics)
- Ansys (FEA)
- Structural Analysis
- Stress and Load Analysis
- Aircraft Structural Fundamentals

LANGUAGES

- English, Marathi, Hindi

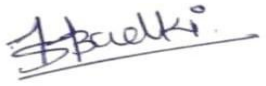
CERTIFICATIONS

- **AutoCAD** (Udemy)
- **CATIA V5** (Udemy)
- **Solidworks** (Udemy)
- **Python** (AICTE approved)

Declaration
I hereby declare that the information provided above is true to the best of my knowledge and belief.

Place: Pune

Signature:



Name: Harshali Suresh Badki